

Abhinav Mahajan

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EDUCATION

Carnegie Mellon University

Master of Science in Computer Vision | GPA: 4.33/4.0

Pittsburgh, PA

December 2026

International Institute of Information Technology (IIIT Bangalore)

(Dual Degree) Master and Bachelor of Technology in Electronics and Communication Engineering

Bangalore, India

July 2025

GPA: 3.63/4.00; Dean's Merit List; Teaching Assistant for courses Machine Learning (AI 511), Visual Recognition (AI 825)

SKILLS

Programming Languages: Advanced: Python; Basic: C++, Java, C, Javascript, Matlab

Libraries: Advanced: PyTorch, NumPy, Pandas, Diffusers, OpenCV, Transformers, trl; Basic: Tensorflow

Specialized: Diffusion Models, LLMs, Multimodal Models, Neural Radiance Fields, Gaussian Splatting, AWS, Docker

PROFESSIONAL EXPERIENCE

Amazon

Bangalore, India

Applied Scientist Intern [Advisor: Arindam Sarkar, Dr. Prakash Mandayam Comar]

January 2025 - June 2025

- Achieved 3.2× top-K recall improvement in next-purchase prediction via MERIT, a novel exposure-mitigated training objective for autoregressive generative XMC with adaptive patching enabling 10× token compression.
- Distilled propensity-aligned embeddings from the trained generative model into lightweight dual-encoder system, enabling bi-directional ranking of customers and categories for personalized ad targeting at scale.
- System achieved +0.26% conversion lift in A/B tests; Paper under review at KDD 2026; Received full-time offer.

Adobe Research

Bangalore, India

Research Intern [Advisor: Dr. KJ Joseph, Dr. Balaji Vasan Srinivasan]

May 2024 - July 2024

- Developed Components-to-Design framework for harmonizing multimodal design assets (images, text, shapes) into aesthetic graphic designs (digital Posters/Flyers) using Large Multimodal Models and diffusion models.
- Designed a training-free identity-preserving composition method through cross-attention and self-attention matrix manipulation for LMM's, significantly improving foreground identity retention while maintaining semantic coherence.
- Paper under review at ICML 2026; Patent in provisional stage; Received full-time offer as MLE-1 in Firefly team.

Adobe Research

Bangalore, India

Research Intern [Advisor: Dr. KJ Joseph, Dr. Balaji Vasan Srinivasan]

May 2023 - August 2023

- Trained a self-supervised graphic design aesthetics scoring model using Siamese networks and contrastive learning to eliminate human annotation bias, achieving 94.97% accuracy in ranking graphic designs.
- Developed a novel genetic algorithm for layout refinement optimizing on the aesthetics scorer, outperforming all baselines; System deployed in Adobe Express; Paper accepted at WACV 2025 (Round 1, 13% acceptance rate).

PUBLICATIONS

- [1] **Abhinav Mahajan*** et al. *MERIT: Mitigating Exposure Bias in Generative XMC for User Propensity*, **Under Review**
- [2] **Abhinav Mahajan***, Abhikhya Tripathi* et al. *Make-It-Pretty: Generating Designs from its Components*, **Under Review**
- [3] **Abhinav Mahajan***, S Goyal* et al. *Design-o-meter: Towards Evaluating and Refining Graphic Designs*, **WACV 2025**
- [4] Ali Abedi*, Tracey JF Colella*, **Abhinav Mahajan** et al. *AVA: AI-driven Virtual Rehabilitation Assistant*, **WCISVR 2023**

PROJECTS

Wiggle and Go! Deformable Object Rapid Motor Adaptation

October 2025 - Present

- Developed two-stage pipeline: adaptation policy estimates unknown rope parameters from safe exploratory motion ("wiggles"), informing trajectory model for accurate goal-conditioned manipulation; preparing RSS 2026 submission.

Hierarchical Activity Analysis of Users

August 2024 - October 2024

- Awarded the MITACS Fellowship and interned at Ontario Tech University, Canada. Developed a platform to crowd source a dataset, contributed to research on screenshot parsing with Vision Language Models and Active-Learning.

Pothole Detection and Severity Estimation

December 2022 - January 2023

- Won the SDAIA Smartathon Hackathon among 1,000+ submissions by training a Mask R-CNN model achieving 82% segmentation accuracy on video data and a 3D reconstruction pipeline of potholes for severity and risk assessment.